



Installation Instruction

Installation Instruction Number: 4021494

Released Date: 15-apr-2004

ISX, Signature, and QSX15 Upgrade Kit Reminders

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Purpose

These instructions detail the major differences between the single-piece and articulated pistons for ISX, Signature, and QSX15 engines. The instructions also highlight the procedures that **must** be followed for a clean and effective rebuild. This is **not** a step-by-step procedure to rebuild the engine. The complete rebuild instructions are detailed in various sections of the Troubleshooting and Repair Manual ISX, Signature, and QSX15 Engines, Bulletin 3666239 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666239.html>).

Note : Due to publication restrictions, the order in which the information is presented below is **not** the order in which the technician will conduct the rebuild.

When rebuilding an engine with single-piece pistons that was originally built with articulated pistons, stamp the dataplate with the letters SP. This will alert future repairing locations that this engine has been rebuilt with single-piece pistons.

For optimum cooling, install a 103 kPa [15 psi] radiator cap and re-calibrate the engine with a November 2003 or later CD-Rom. When the rebuild is complete, refer to Procedure 014-003 (</qs3/pubsys2/xml/en/procedures/10/10-014-003.html>) in the Troubleshooting and Repair Manual ISX, Signature, QSX15 Engines, Bulletin 3666239 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666239.html>) for proper engine run-in.

CPL	Application	ISX and QSX Upgrade Kit Part Number
2440, 2441, 2627, 2632, 2803, 2814, 2933, 8048, 8083, 8242, 8244, 8255, 2857, 8258	ISX and ISX2	4089494
2442, 2483, 2630, 2633, 2801, 2816, 2836, 2900, 8246	ISX3 and Signature	4089495
2628	ISX 400 and 450	4089494
2629	ISX 500, 500 ST2, 475, and 475 ST2	4089494
2631	ISX 450	4089494
2834	Signature 600 and 565	4089495
8046	ISX 450 ST2	4089494

CPL	Application	ISX and QSX Upgrade Kit Part Number
8047	ISX 400	4089494
8124, 8281, and 8285	ISX2 and ISX3 EGR	4089497
Description		Part Number (Quantity)
Overhaul Conversion Kit / Piston		4089494 (1) / 4920308 (6)
Overhaul Conversion Kit / Piston		4089495 (1) / 4929309 (6)
Overhaul Conversion Kit / Piston		4089496 (1) / 4920307 (6)
Overhaul Conversion Kit / Piston		4089497 (1) / 4920309 (6)
Retaining Ring		3064305 (12)
Lower Engine Gasket Set		3102234 (1)
O-ring Seal		3678798 (6)
Main Bearing Set		3800298(1)
Bypass Valve Plunger		4024300 (1)
Upper Engine Gasket Set		4059363 (6)
Connecting Rod		4059429 (6)
Ring Set		4089406 (6)
Cylinder Liner		4101507 (6)
Rod Bearing Set		4089405 (6)

Related Procedures

10-001-054-topic (/qs3/pubsys2/xml/en/procedures/10/10-001-054-topic.html)	Piston and Connecting Rod Assembly (/qs3/pubsys2/xml/en/procedures/10/10-001-054-topic.html)
10-002-004-topic (/qs3/pubsys2/xml/en/procedures/10/10-002-004-topic.html)	Cylinder Head (/qs3/pubsys2/xml/en/procedures/10/10-002-004-topic.html)

Last Modified: 29-Apr-2004
